

The Human Eye and the Colourful World (Set B) — MCQ Worksheet

Science • Chapter 10 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The lens of the human eye is:

- (A) Convex with adjustable focal length
- (C) Plane

- (B) Concave
- (D) Fixed focal length

Q2. Image on retina is:

- (A) Real and inverted
- (C) Real and erect

- (B) Virtual and erect
- (D) Virtual and inverted

Q3. Pupil controls:

- (A) Amount of light entering eye
- (C) Vision in dark only

- (B) Focusing
- (D) Sound perception

Q4. Cornea provides most of the eye's:

- (A) Refraction (about 2/3 of total)
- (C) Sensitivity to light

- (B) Pigmentation
- (D) Stability

Q5. Power of accommodation of the eye is due to:

- (A) Ciliary muscles changing lens shape
- (C) Retina moving

- (B) Pupil contracting
- (D) Iris changing colour

Q6. Far point of normal eye is at:

- (A) Infinity
- (C) 1 m

- (B) 25 cm
- (D) 10 m

Q7. Near point of normal eye is at:

- (A) 25 cm
- (C) 10 cm

- (B) Infinity
- (D) 1 m

Q8. Hypermetropia (long-sight) is corrected with:

- (A) Convex lens
- (C) Cylindrical lens

- (B) Concave lens
- (D) Plane lens

Q9. Myopia (short-sight) is corrected with:

- (A) Concave lens
- (C) Cylindrical

- (B) Convex lens
- (D) Plano-convex

Q10. Splitting of white light into seven colours is:

- (A) Dispersion
- (C) Polarisation

- (B) Diffraction
- (D) Refraction only

Q11. Red light bends:

- (A) Least through prism
- (C) Equally

- (B) Most
- (D) Not at all

Q12. Violet light bends:

- (A) Most
- (C) Equally

- (B) Least
- (D) Negligibly

Q13. Blue colour of clear sky is due to:

- (A) Scattering of light by air molecules
- (C) Refraction by air

- (B) Reflection from clouds
- (D) Dispersion

Q14. Sun appears reddish at sunrise/sunset because:

- (A) Longer path; shorter wavelengths scattered out
- (C) Earth tilts

- (B) It is hotter
- (D) It is closer

Q15. Rainbow forms due to:

(A) Dispersion and total internal reflection of light by raindrops

(C) Refraction only

(B) Reflection only

(D) Absorption

Q16. Why do stars twinkle but not planets?

(A) Stars are point sources; atmospheric refraction causes twinkling. Planets are extended sources.

(C) Planets have own light

(B) Stars are dim

(D) Planets are stationary

Q17. Tyndall effect is:

(A) Scattering of light by colloidal particles

(C) Total internal reflection

(B) Dispersion in prism

(D) Reflection in mirror

Q18. Why is the colour of danger signals red?

(A) Red has longest visible wavelength; scattered least so visible from far in fog

(C) Tradition only

(B) Red is rare

(D) Cheaper

Q19. Power of a lens for far vision in myopic person whose far point is at 1 m is:

(A) -1 D

(C) -5 D

(B) +1 D

(D) +5 D

Q20. A hypermetropic person has near point at 1 m. The power of lens to see object at 25 cm is approximately:

(A) +3 D ($1/0.25 - 1/1 = 3$)

(C) +4 D

(B) +1 D

(D) -3 D

Q21. Why does a small drop of water on a paper magnify the print under it?

(A) The drop acts as a convex lens

(C) Static electricity

(B) Water bends paper

(D) Surface tension only

Q22. If a person can read fine print only when book is held very near, the defect is:

(A) Myopia (short-sightedness)

(C) Presbyopia

(B) Hypermetropia

(D) Astigmatism

Q23. The bluish colour of sea water near horizon is due to:

(A) Scattering of light

(C) Both A and B contribute

(B) Reflection of sky

(D) Refraction

Q24. Atmospheric refraction makes Sun appear:

(A) Visible for about 2 minutes before actual sunrise and after sunset

(C) Yellow always

(B) Larger than it is

(D) Stationary

Q25. Cone cells in retina enable:

(A) Colour vision in bright light

(C) Hearing

(B) Vision in dim light

(D) Smell